AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #11

Activity Name: UART Programming

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**02|10|2026**
Submission Date:	**02|10|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to drive UART module. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
#include <LPC21xx.H>

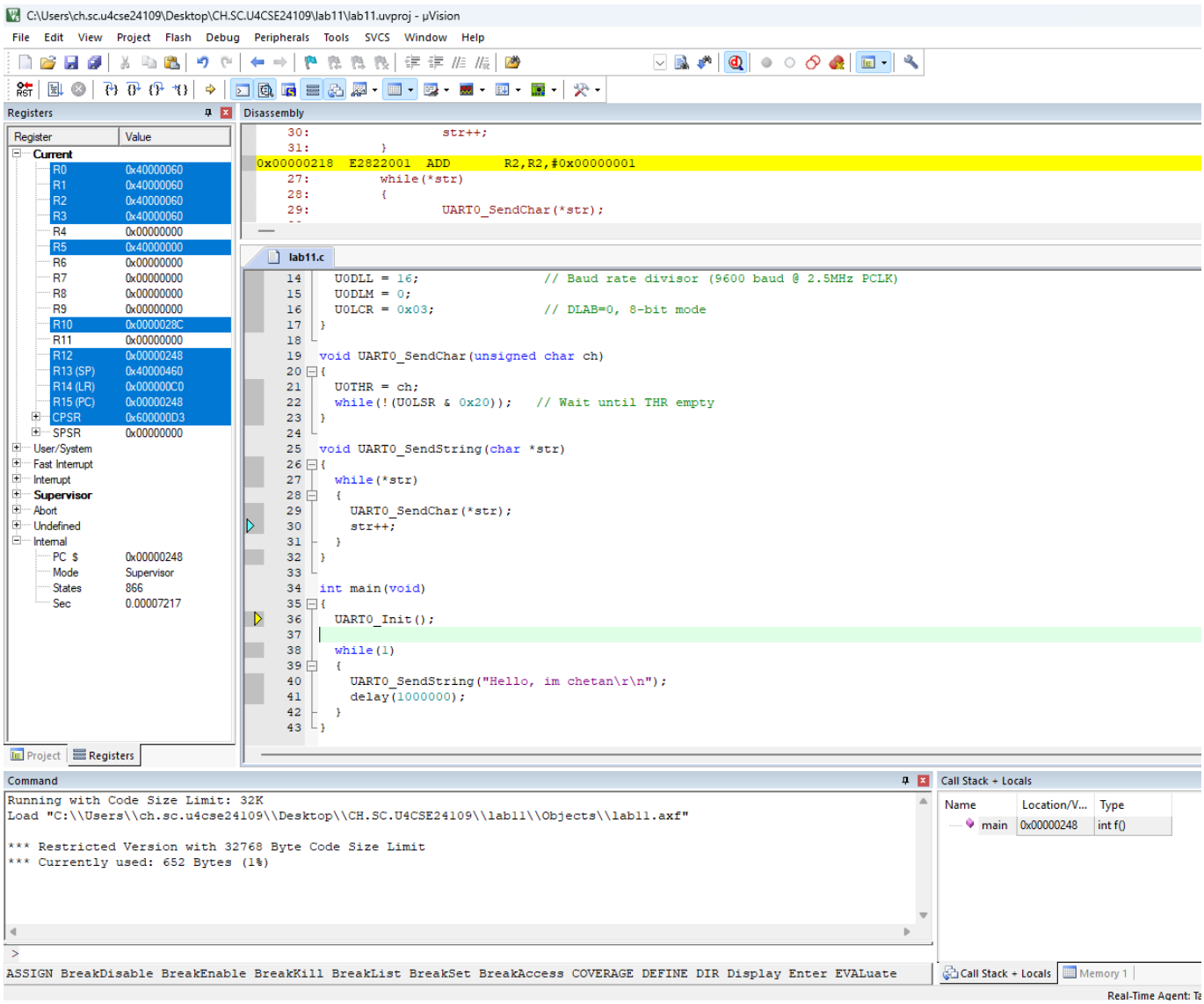
void delay(unsigned int t)
{
unsigned int i;
for(i=0; i<t; i++);
}
void UART0_Init(void)
{
PINSEL0 |= 0x00000005;   // P0.0 = TXD0, P0.1 = RXD0

U0LCR = 0x83;             // 8 bits, 1 stop, no parity, DLAB=1
U0DLL = 16;                 // Baud rate divisor (9600 baud @
2.5MHz PCLK)
U0DLM = 0;
U0LCR = 0x03;              // DLAB=0, 8-bit mode
}
void UART0_SendChar(unsigned char ch)
{
U0THR = ch;
while(!(U0LSR & 0x20));   // Wait until THR empty
}
void UART0_SendString(char *str)
{
while(*str)
{
UART0_SendChar(*str);
str++;
}
}

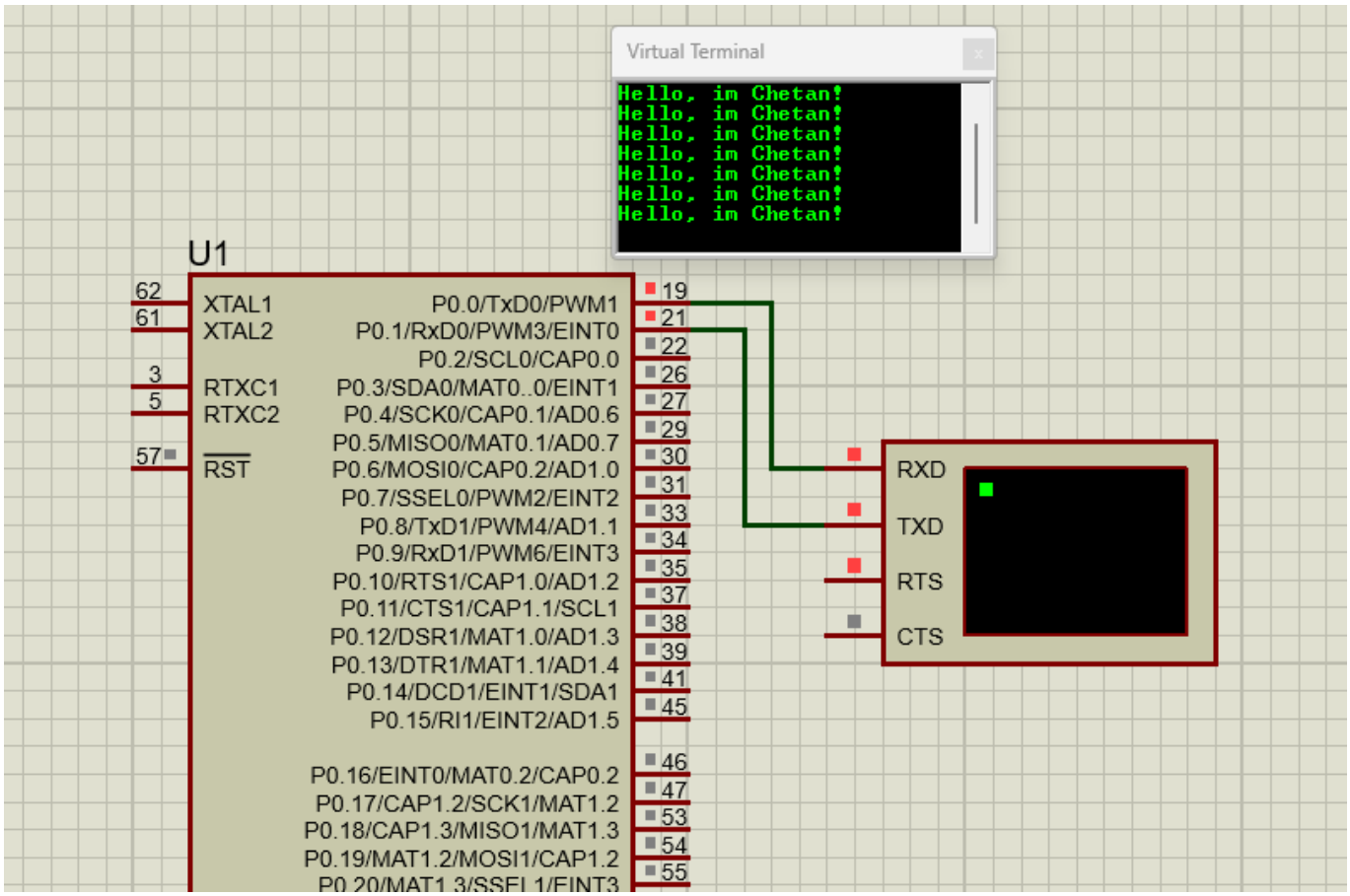
int main(void)
{
UART0_Init();

while(1)
{
UART0_SendString("Hello, im chetan\r\n");
delay(1000000);
}
}
```

OUTPUT WINDOW:



OUTPUT:


Virtual Terminal
Hello, im Chetan!
Hello, im Chetan!
Hello, im Chetan!
Hello, im Chetan!
Hello, im Chetan!
Hello, im Chetan!
Hello, im Chetan!
U1
62 XTAL1
61 XTAL2
3 RTXC1
5 RTXC2
57 RST
P0.0/TxD0/PWM1 19
P0.1/RxD0/PWM3/EINT0 21
P0.2/SCL0/CAP0.0 22
P0.3/SDA0/MAT0..0/EINT1 26
P0.4/SCK0/CAP0.1/AD0.6 27
P0.5/MISO0/MAT0.1/AD0.7 29
P0.6/MOSI0/CAP0.2/AD1.0 30
P0.7/SSEL0/PWM2/EINT2 31
P0.8/TxD1/PWM4/AD1.1 33
P0.9/RxD1/PWM6/EINT3 34
P0.10/RTS1/CAP1.0/AD1.2 35
P0.11/CTS1/CAP1.1/SCL1 37
P0.12/DSR1/MAT1.0/AD1.3 38
P0.13/DTR1/MAT1.1/AD1.4 39
P0.14/DCD1/EINT1/SDA1 41
P0.15/RI1/EINT2/AD1.5 45
P0.16/EINT0/MAT0.2/CAP0.2 46
P0.17/CAP1.2/SCK1/MAT1.2 47
P0.18/CAP1.3/MISO1/MAT1.3 53
P0.19/MAT1.2/MOSI1/CAP1.2 54
P0.20/MAT1.3/SSEL1/EINT3 55
RXD
TXD
RTS
CTS